

Software Requirements Specification

Yard allocation, crane sequencing and gate slot requirements with performance acceptance criteria

Deliverable 2 of 6 · Case Study no. 39 · PortFlow

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Case	PortFlow - Container Terminal Yard and Gate Operations		

1. Introduction

1.1 Purpose

This SRS specifies the requirements for **PortFlow**, a system that plans container yard positions, sequences crane moves and appoints truck arrival slots for a terminal handling 1.4 million TEU a year, and executes the plan in real time.

1.2 Problem background

- Trucks queue at the gate for an average of **71 minutes**.
- Yard cranes make unnecessary re-handles because containers are stacked without regard to departure order: **1.8 moves per container** against a target of **1.25**.
- The berth plan is made on a whiteboard.

1.3 Stakeholders

Stakeholder	Interest
Terminal planners	Yard and berth plans, re-plan on change
Crane / RTG operators (union)	Clear job lists, safe sequences, fair slot rules
Gate staff	Fast validation, fewer queues
Trucking companies	Predictable slots, short waits
Port authority	Concession performance guarantee
Shipping lines	Vessel turnaround

2. Overall Description

- Planning layer:** yard planner, crane sequencer, slot service, berth plan importer.
- Execution layer:** real-time controller, gate and crane adapters. It continues running if the planning layer fails.
- Constraints:** legacy crane/PLC interfaces; union consultation on slot rules; concession deadline of 34 weeks.
- Assumptions:** historical vessel, gate and yard data of at least 3 months is available for replay testing.

3. Functional Requirements

3.1 Yard allocation rules (YA)

ID	Requirement	Priority
YA-1	The system shall assign every container a yard position based on its expected departure time (vessel ETD or truck pickup window).	Must
YA-2	A container leaving earlier shall not be placed beneath one leaving later in the same stack.	Must
YA-3	Allocation shall consider container type (reefer, hazardous, empty, 20/40 ft), weight class and discharge port.	Must
YA-4	When an ETD changes, the system shall re-plan and flag containers that become blocked.	Must
YA-5	Stack height limits and reefer-plug availability shall be enforced.	Must

3.2 Crane sequencing rules (CS)

ID	Requirement	Priority
CS-1	The system shall generate an ordered job list per crane that minimises total moves and travel.	Must
CS-2	Where a re-handle is unavoidable, the destination slot shall be chosen so it does not cause a further re-handle.	Must
CS-3	Sequences shall respect vessel load order and safety rules.	Must
CS-4	Sequences shall be re-optimised when a crane fails or priorities change.	Should

3.3 Gate slot appointment rules (SA)

ID	Requirement	Priority
SA-1	Trucking companies shall book a slot (container ID, time window) through a web portal or API.	Must
SA-2	Slots shall be limited by gate capacity and yard/crane capacity for that window (no overbooking).	Must
SA-3	If the requested slot is full, the system shall offer alternatives and send confirmations, reminders and changes.	Must
SA-4	Early or late trucks shall be handled by a configurable grace window with a rebooking path.	Should
SA-5	Slot rules shall be configurable and communicated before changes take effect (union and carrier consultation).	Should

3.4 Berth plan (BP) and execution (EX)

ID	Requirement	Priority
BP-1	The berth plan (vessel ETA/ETD, berth, crane allocation) shall be digital and shared with yard planning, replacing the whiteboard.	Must
EX-1	The execution layer shall issue jobs to cranes, validate trucks at the gate and record actual moves in real time.	Must
EX-2	If the planning engine fails, execution shall continue using the last published plan and fallback rules.	Must

4. Non-Functional Requirements

ID	Requirement
NFR-1	Plan regeneration after a change shall take at most 5 minutes (proposed, to be agreed with the terminal).
NFR-2	The execution layer shall be available at least 99.9%, independent of the planning engine.
NFR-3	A gate validation transaction shall complete within 10 seconds.
NFR-4	Every move and every plan version shall be logged for audit.
NFR-5	Role-based access; carriers see only their own bookings.
NFR-6	Integration with existing crane/PLC equipment through adapters.

5. Performance Acceptance Criteria: the Re-handle Metric

5.1 Measurement definition

Re-handle ratio = total crane lifts on container retrievals ÷ number of containers retrieved

An ideal retrieval needs exactly **1.0** move. A move is counted only when a container is physically lifted. The ratio is measured for outbound containers (to vessel or gate) over a defined period.

Current: 1.8. Target: ≤ 1.25 .

5.2 Acceptance tests

ID	Acceptance criterion
AC-0 (calibration)	Replaying historical data through the legacy rules shall reproduce a ratio of 1.8 ± 0.1 . This proves the replay is realistic and must pass first.
AC-1 (target)	Replaying the same data through PortFlow shall give an overall ratio ≤ 1.25 and a peak-week ratio ≤ 1.35 (peak tolerance proposed, to be agreed).
AC-2 (gate)	Average gate turnaround shall fall from 71 minutes to a target agreed with the port authority before build (the case study does not state a figure).

6. Traceability

YA-1..5 and CS-1..4 → AC-0/AC-1 (re-handle target). SA-1..5 → AC-2 (gate wait). EX-2 → simulation test SIM-7 (planning failure). Test cases are defined in the Test Design document.